🔍 Image Classification Pipeline Using Moment Features, PCA/ICA, and k-NN

This repository contains MATLAB scripts for a complete pipeline to classify grayscale imagesusing image preprocessing, feature extraction (Legendre Moments, PCA, ICA), and classification via the k-Nearest Neighbors (k-NN) algorithm.

🧩 Pipeline Overview

The pipeline consists of **four main stages**, each handled by a separate script:

---

1. Image Preprocessing

Script: `image_preprocessing.m`
Purpose:

- Apply a global threshold to grayscale images to reduce intraclass variance
- Divide each image into 25 equal-sized sub-images (5×5 grid)
- Save each sub-image individually for feature extraction

---

2. Feature Extraction – Legendre Moments

Script: `extract_legendre_moments.m` and `LM.m` **Purpose:**

- Compute Legendre Moments up to user-defined orders
- Save feature vectors with appended class labels to `.mat` files

---

3. Feature Extraction – PCA/ICA

Script: `pca_ica_feature_extraction.m`
Purpose:

- Compute PCA and ICA
- Apply:
  - **PCA** to retain all variance or 95% variance
  - **ICA** to extract 100 independent components
- Save PCA scores, ICA mixing matrix, and variance information

---

4. Classification – k-NN (Varying Features)

Script: `knn_classification_features.m`
Purpose:

- Perform k-NN classification on ICA, PCA, and LM-derived features
- Iterate over different **numbers of features**
- Output performance metrics:
  - Accuracy
  - Sensitivity
  - Specificity
  - Classification Loss
  - Mean Squared Error (MSE)

- **Generates Figure 4**

---

5. Classification – k-NN (Varying Training Size)

Script: `knn_classification_trainingsize.m`
Purpose:

- Perform k-NN classification using different **training sizes**
- Output same performance metrics as above
- Plot **ROC curves at 80% training size**
- **Generates Figures 5 and 6**

Output Summary

- Sub-images (`.jpg`)
- Legendre Moments, PCA, and ICA features
- Classification performance metrics
- Plots (ROC curves, accuracy trends)

Suggested Workflow

1. **Preprocess Images**
   `→` Run `image_preprocessing.m`
2. **Extract Features (Legendre, PCA, ICA)**
   `→` Run `extract_legendre_moments.m` + `LM.m` and `pca_ica_feature_extraction.m`
3. **Classify & Evaluate**
   `→` Run `knn_classification_features.m` and `knn_classification_trainingsize.m`

Requirements

- MATLAB (R2020a or later recommended)
- Image Processing Toolbox
- Custom Legendre Moment functions (`LM.m`)